WORKED SOLUTION

STEP II 2009, Question 1

Pure • Symmetry of Curves

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s2-q1

The key idea

Almost nothing here needs computing: symmetry does the work. Both curves are unchanged by reflection in the lines $y = x$ and $y = -x$, so once we have one intersection point, the others come from reflecting it. The rectangle then follows from the shape of the four points. For the area, we never need to find α and β themselves: the answer turns out to need only one combination of them, and we can get that straight from $\alpha^4 + \beta^4 = u$ and $\alpha\beta = v$.

Solution

Lines of symmetry

A curve is symmetric in a line if reflecting any point of the curve in that line gives another point of the curve. The four reflections we need are

$$(x,y) \mapsto (-x,y), \quad (x,y) \mapsto (x,-y), \quad (x,y) \mapsto (y,x), \quad (x,y) \mapsto (-y,-x),$$

which are the reflections in $x = 0$, $y = 0$, $y = x$ and $y = -x$.

The curve $x^4 + y^4 = u$. The expression $x^4 + y^4$ is unchanged by all four maps, because fourth powers ignore signs and the expression is symmetric in x and y. So this curve has four lines of symmetry:

$$x = 0, \qquad y = 0, \qquad y = x, \qquad y = -x.$$

The curve $xy = v$. Swapping x and y leaves xy unchanged, and so does $(x,y) \mapsto (-y,-x)$, since $(-y)(-x) = xy$. Changing the sign of just one coordinate sends xy to $-xy$, and since $v > 0$ the point $(-x,y)$ is then on $xy = -v$, not on our curve. So this rectangular hyperbola has exactly two lines of symmetry:

$$y = x, \qquad y = -x.$$

The four intersection points

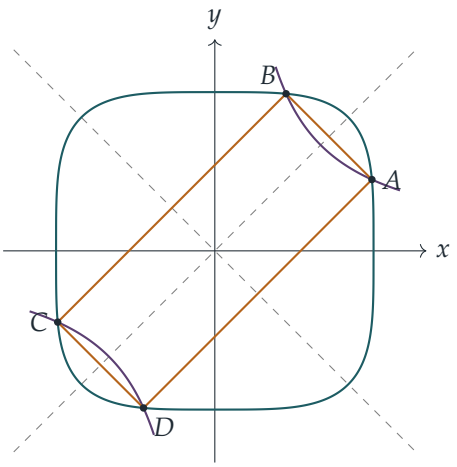
Let S be the reflection in $y = x$ and T the reflection in $y = -x$. Each maps both curves to themselves, so each maps an intersection point to an intersection point. Starting from $A = (\alpha, \beta)$ we get

$$S(A) = (\beta, \alpha), \qquad T(A) = (-\beta, -\alpha), \qquad T(S(A)) = (-\alpha, -\beta).$$

These four points are distinct, because $\alpha > \beta > 0$. There are no other intersections. Substituting $y = v/x$ into the quartic gives $x^8 - ux^4 + v^4 = 0$, a quadratic in x^4 with at most two positive roots, so there are at most four real values of x.

To label them anticlockwise, look at their polar angles. Since $0 < \beta < \alpha$, the point A has angle θ with $0 < \theta < 45°$. Then (β, α) has angle $90° - \theta$, $(-\alpha, -\beta)$ has angle $180° + \theta$ and $(-\beta, -\alpha)$ has angle $270° - \theta$. These angles increase in that order, so

$$B = (\beta, \alpha), \qquad C = (-\alpha, -\beta), \qquad D = (-\beta, -\alpha).$$



(The picture shows the case $u = 81$, $v = 4$: the quartic in teal, the hyperbola in plum.)

$ABCD$ is a rectangle

The diagonals are AC and BD. The midpoint of AC is $(\frac{\alpha-\alpha}{2}, \frac{\beta-\beta}{2}) = (0, 0)$, and likewise the midpoint of BD is $(0, 0)$. So the diagonals bisect each other, and $ABCD$ is a parallelogram. Their lengths are

$$AC = \sqrt{(2\alpha)^2 + (2\beta)^2} = 2\sqrt{\alpha^2 + \beta^2}, \qquad BD = \sqrt{(2\beta)^2 + (2\alpha)^2} = 2\sqrt{\alpha^2 + \beta^2},$$

which are equal. A parallelogram with equal diagonals is a rectangle, so $ABCD$ is a rectangle.

Ask yourself Why does a parallelogram with equal diagonals have to be a rectangle?

All four vertices are at the same distance $\sqrt{\alpha^2+\beta^2}$ from O, so they lie on a circle with centre O, and AC and BD are diameters of it. By the angle in a semicircle, each angle of $ABCD$ is a right angle. To check directly instead, $\overrightarrow{AB} = (\beta-\alpha,\ \alpha-\beta)$ and $\overrightarrow{BC} = (-\alpha-\beta,\ -\alpha-\beta)$, with scalar product

$$-(\beta-\alpha)(\alpha+\beta)-(\alpha-\beta)(\alpha+\beta) = (\alpha+\beta)[(\alpha-\beta)-(\alpha-\beta)] = 0.$$

The area

The side lengths are

$$AB = \sqrt{2(\alpha-\beta)^2} = \sqrt{2}\,(\alpha-\beta), \qquad BC = \sqrt{2(\alpha+\beta)^2} = \sqrt{2}\,(\alpha+\beta),$$

using $\alpha > \beta > 0$ to take the positive square roots. So

$$\text{area} = AB \cdot BC = 2(\alpha-\beta)(\alpha+\beta) = 2(\alpha^2-\beta^2).$$

Now use the fact that A lies on both curves, so $\alpha^4+\beta^4 = u$ and $\alpha\beta = v$. Then

$$(\alpha^2-\beta^2)^2 = \alpha^4+\beta^4-2\alpha^2\beta^2 = u-2v^2.$$

Since $\alpha^2-\beta^2 > 0$, we take the positive root, and

$$\boxed{\text{area of } ABCD = 2\sqrt{u-2v^2}}$$

This also shows that the question's assumption of four distinct points forces $u > 2v^2$. If $u = 2v^2$ then $\alpha = \beta$ and the curves only touch.

For $u = 81$ and $v = 4$: $u - 2v^2 = 81 - 32 = 49$, so the area is $2 \times 7 = 14$, as required.

Checks

For $u = 81$, $v = 4$, solving $x^8 - 81x^4 + 256 = 0$ numerically gives $\alpha \approx 2.9690$ and $\beta = 4/\alpha \approx 1.3472$. Then $2(\alpha^2-\beta^2) \approx 14.000$, and the shoelace formula on the four vertices gives the same area. As u falls towards $2v^2$ the two points in each quadrant merge on $y = x$ and the rectangle flattens to a segment, matching area $2\sqrt{u-2v^2} \to 0$.

Common slips

- Giving the coordinate axes as lines of symmetry of $xy = v$. Reflecting in an axis sends the branch in the first quadrant to the second quadrant, where $xy < 0$.
- Labelling the points in the wrong order, for example taking $B = (-\alpha, -\beta)$. Then $ABCD$ is traversed as a crossed quadrilateral.
- Trying to solve for α and β explicitly. The area needs only $\alpha^2-\beta^2$, which comes straight from u and v.

Beyond STEP

The four reflections and rotations that preserve $x^4 + y^4 = u$ form the symmetry group of a square, the dihedral group of order 8. The hyperbola keeps only a subgroup of order 4 (two reflections, the half-turn and the identity). Intersection points are permuted by the symmetries the two curves share, and finding objects by the action of a group like this is a theme of first-year group theory. Allenby's *Rings, Fields and Groups* is a good place to meet these ideas.

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